

Thank you very much for sending your manuscript "Quantity discrimination, decision-making, and the role of early- life conditions in a lizard" to Evolutionary Ecology. Given the unique review history of this manuscript, I decided to make decision without sending it out to an external review. My own reading suggests that this is a carefully executed experimental research and the result is relevant for a broad range of research topics that our journal covers. I have some comments, though, which I summarize below. All are very minor in nature, but one highlight is that at times the authors conflate the lack of evidence to the evidence of absence. My suggestion is, rather than saying "our results suggest lizards cannot count", write in the tone of "our results are inconsistent with the idea that lizards can count", throughout the manuscript. Please see my suggestions below.

All in all, I think the manuscript is ready for publication in our journal pending minor revision of the items I summarize in my comments attach to this decision letter.

Response: Thank you very much for your understanding and all your comments. We address your concerns below.

Reviewer 1

Line 43: Please either avoid an abbreviation (GCs) in abstract, or if used explain in full.

Response: We have replaced "GCs" with "glucocorticoids"

Line 47-50: The message here is too generic to be useful. I guess the authors here imply that the null result is still interesting either because (a) the lack of evidence is an interesting biological signal in need of explanation, or (b) the study clarified what needs to be done in the future to detect the effect if you take a position that your result reflects a lack of power to detect an effect. Some specific arguments along these lines (doesn't have to be these two) would be tremendously useful because they specify your novel contribution.

Response: Thank you for your feedback. We have included more details about the results on those lines: "Contrary to our predictions, lizards did not show clear preferences toward the larger items, providing no evidence of spontaneous quantity discrimination in *L. guichenoti*. Furthermore, prenatal conditions did not influence latency to make a choice, also a proxy for decision-making." (Lines 47-50).

Line 71: probably "why different methods can..." or "why methods matter." would reads more smoothly?

Response: We have modified the sentence accordingly to the reviewer's suggestion: "Numerous reasons exist for why different methods can lead to different results" (Line 72)

Line 95: what is "cumulative surface area"? Could you explain that briefly here so readers outside of the field can understand your argument?

Response: We thank the reviewer for this suggestion. Cumulative surface area refers to the total area occupied by a set of items combined and is considered one common continuous quantity cues animals could use (Agrillo and Bisazza 2014, *J. Neurosci. Methods*). We have added a brief clarification of this term in the main text: "In spontaneous choice tests, for instance, animals may rely on aspects such as the total area occupied by a set of items combined—cumulative surface area (ANS-related)—rather than numerical (OFS-associated) differences when selecting between two sets of food items (Agrillo and Bisazza 2014)." (Lines 95-98).

Line 103: "competent in" -> "capable of" to make it sounds more neutral?

Response: We do agree with the reviewer, we have made the suggested modification.

Line 193-194: is there particular reason for this timing during development was chosen? There are so much happening during the embryonic stage, so exactly what temporal window you do the treatment may have enormous impact on results.

Response: We agree that the temporal window employed can significantly impact the results. Prenatal stages were selected for two reasons: first, oviparous animals cannot avoid the conditions they experience during embryonic development (i.e. eggs cannot move); second, glucocorticoids are likely metabolised and cleared within the egg, meaning that postnatal manipulations may not be as ecologically relevant as manipulations immediately after laying, which more closely simulates the natural situation in which females deposit glucocorticoids into their eggs. We have included a brief sentence here in this regard: "We manipulated prenatal conditions for two reasons: first, oviparous species, like *L. guichenoti*, may be particularly susceptible to the environment in which embryos develop, given their inability to relocate to more favourable conditions during this stage. Second, GC transmission from mothers is likely more ecologically relevant at these early stages, as glucocorticoids may be metabolised later during egg development (Carter et al., 2018)." (Lines 196-200).

Line 264-266: do you need this sentence?

Response: We agree with the reviewer that, since no animals were removed under that criterium, this sentence may not be necessary. We have removed the sentence.

Line 313: According to Table S3, HDPI of the effect of sex is [-0.82, -0.10]. For here and elsewhere, it would be a good idea to cross-check those numbers are consistent between supplementary tables and in the main text.

Response: Thank you very much for your attention to detail. Upon review, we identified the source of this discrepancy: the main text reported highest density posterior intervals (HDPIs), while the supplementary tables reported standard 95% credible intervals. We have now clarified this distinction in all supplementary tables.

Line 329-332: The lack of evidence is not the evidence of absence. Something like “Thus, our results were not consistent with the hypotheses that lizards used numerical cues when making foraging decisions or that early environmental conditions affected such decision-making. More generally, we found no evidence that *L. guichenoti* relied on numerical or other quantitative cues under the conditions tested.” would work better.

Response: We agree with the reviewer that caution should be taken with null results. We have modified our sentence to match what is suggested here: “Our results were not consistent with the hypothesis that lizards discriminate between different numbers of crickets. Similarly, we did not see lizards’ performance to vary under our prenatal experimental conditions.” (Lines 333-335)

Line 354-355: For the same reason as above, I’d be more cautions about this interpretation. “our results may reflect...”, for instance.

Response: Thank you for this suggestion. We have modified the sentence as suggested: “One potential explanation for our results is that *L. guichenoti* relies on the ANS for quantity discrimination.” (Lines 354-356).

Line 382: teh -> the

Response: Thanks for the attention to detail. We have corrected this.

Line 404-418: Interesting discussion. I see a possibility to mention the timing of treatment here. To my understanding, brain development starts right after the phylotypic stage when other organs also start to grow, then the first phase of development ends before they give birth (except for some mammals and altricial birds), then the second phase continue to the adulthood. From the viewpoint of experimentally affecting key neuronal functions, it is probably critical to target the timing during the first phase here, because in the second, phase, brain is just growing larger and larger with the same organizational feature, and before that, brain has not been completely differentiated yet.

Response: We appreciate the reviewer's suggestion and we have included some sentence in this regard: "Additionally, future studies should focus on the effect of environment at different times of early life targeting specific phases of neural and brain development. Here, we manipulated prenatal conditions because we considered that this developmental stage would be the most influenced by temperature and maternal transmitted GCs (see Manipulating Early Thermal and CORT Environments above). However, examining the effects of these or other environmental factors at more advanced stages of development could yield different results."

We appreciate the reviewer's insightful suggestion and have added the following to the discussion: "Additionally, future studies should consider the timing of environmental manipulations relative to specific phases of neural development. Here, we targeted prenatal conditions as we considered this stage to be most susceptible to the effects of temperature and maternally transmitted GCs (see Manipulating Early Thermal and CORT Environments above). However, examining the effects of these or other environmental factors at more advanced stages of development could yield different results." (Lines 424-428)

Line 422-424: Again, it would be safer to write here along the context of "our results are not consistent with the idea that lizards rely on numerical information for decision making"

Response: Thank you for your feedback. We have modified our sentence to make it more cautious: "Our findings do not align with the hypothesis that *L. guichenoti* relies on numerical information for foraging decisions, which limits the potential for early-life conditions to influence this specific ability." (Lines 432-434)